fastTIGER: A rapid method for estimating evolutionary rates of sites from large datasets

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Abstract— **The evolutionary processes vary among sites of an alignment, called rate heterogeneity, that must be properly handled when analyzing the evolutionary relationships among species based on their genomic data. To this end, methods have been proposed to estimate the relative evolutionary rates between sites. Tree Independent Generation of Evolutionary Rates (TIGER) is a popular method to estimate the evolutionary rates among sites. However, the TIGER method is computationally expensive to calculate the evolutionary rates for large datasets, especially for whole genome datasets. In this paper, we present a simplified, fast, and accurate method, called fastTIGER, to estimate evolutionary rates for large datasets. Experiments on several large real datasets show that the evolutionary rates from the fastTIGER method have a reasonable correlation with ones estimated from the TIGER method while the fastTIGER method is several orders of magnitudes faster than the TIGER method. Moreover, the site rates estimated by fastTIGER method are as good as the ones estimated from the TIGER method in partitioning alignments to build maximum likelihood trees. The fastTIGER method enhances us to study the evolutionary relationships among species using their genomic data.**

Keywords— evolution rates, site-specific rates, TIGER

I. Introduction

A number of studies have proved that the evolutionary rates are heterogeneous among sites [1]–[3]. In coding regions, it is obvious that the evolutionary rates are affected by the selective pressures at sites. In non-coding regions, the rate variation is well discussed in different studies [4], [5]. The failure in representing the rate heterogeneity in models of evolution often leads to an incorrect reconstruction of phylogenies [6]. A simple way to handle the rate heterogeneity problem is to classify sites into different rate categories based on the biological properties of sites. However, the method is only applicable for small and well-studied datasets.

A number of computational methods have been proposed to estimate site rates for a given alignment [7]–[11]. Most of these methods are tree-based methods, i.e., at first, a phylogeny is constructed, then site rates are calculated based on the constructed tree. For example, in the slow-fast method [7], sequences are divided into different taxonomic groups. For a given site, the number of changes at the site is calculated for each individual group and summed up to estimate the rate for the site. Each site is marked as a slow or fast evolving site if the number of changes at the site is less or greater than a predefined threshold. Though it is not required explicitly, slow-fast method need a tree to defined taxonomic group sequences.

The method proposed by Mayrose et. al. [10] infers site rates using a fixed phylogenetic tree or over the entire tree space. In the case of using a fixed tree, the rate for a site is estimated as the expectation of rate over all categories of a discrete approximation of posterior rate distribution. The drawback of this approach is that the initial tree may be incorrect resulting in incorrect estimated rates. Note that the site rates vary when changing topology or branch lengths of the tree. In the case of using the entire tree space, the law of total probability is used to compute site rates. This overcomes the problem of using a fixed tree but using the whole tree space causes a massive computation and cannot be applied to large datasets.

Tree-independent methods have been proposed to estimate site rates without requiring a tree. The TIGER method [8] is the widely-used tree-independent method, that infers evolutionary site rates based on the similarity between sites. Its key idea is that fast-evolving sites should have lower similarities with other sites on average, while slow-evolving sites should have higher similarities with other sites. The TIGER method is less biased than tree-dependent methods in calculating evolutionary site rates and used in a number of studies [12] – [14]. However, examining the TIGER rate method with large genomic datasets showed that it took too long to estimate the similarity between sites for long alignments, for example, the method did not finish after 3 days running on a dataset which has 33 sequences, each sequence has 1541 loci and 539526 sites.

In this paper, we simplify the TIGER method to develop the fastTIGER method to handle large genomic datasets. The fastTIGER method estimates the similarity between two sites by counting the number of sequence pairs having the same states in both sites. Experiments with real large genomic datasets showed that the fastTIGER method is orders of magnitudes faster than the TIGER method, and still procedures equally good results. The rest of the paper is organized as follows: The fastTIGER method will be represented in section II (Methods). Section III (Experiments and Results) will describe the experiments on real genomic datasets and discuss results obtained from both TIGER and fastTIGER methods. We provide discussions, remarks, and recommendations in the last section.

II. Methods

A. The TIGER method

A multiple sequence alignment (MSA) is an alignment of three or more homologous sequences (i.e., sequences evolved from a common ancestral sequence) and can be represented by a matrix. Each row in the matrix represents a sequence

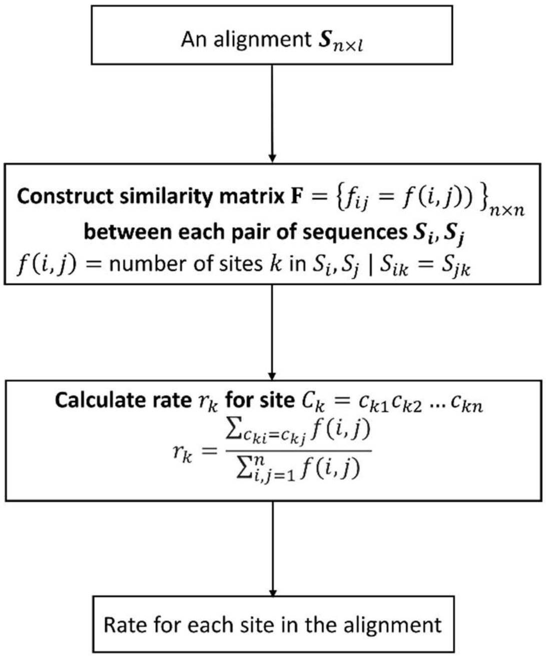

Fig. 1. The flowchart of the fastTIGER method to estimate site rates from an alignment

while each column represents aligned homologous sites. A column of an MSA is called a site or a character, and the number of columns is the length of the MSA. A point mutation in a sequence represents a change of a sequence at a site. There are three kinds of point mutations: substitution, insertion, and deletion. When aligning sequences, deletion is represented by gaps.

Specifically, let $S = (s_{ij})$ be a multiple sequences alignment of n sequences with length l where s_{ij} is the state of sequence i^{th} at site j. The state s_{ij} can be 'A', 'C', 'G', 'T' for nucleotide data or any of twenty amino acid types for amino acid data (s_{ij} can be a gap indicating a deletion). The relative evolutionary rates of sites can be calculated using the similarity between sites in the alignment. The TIGER method calculates site rates by analyzing the similarity of patterns at sites. The method can be used for any data types, i.e., nucleotide, protein, or morphological data. Given an alignment *S*, the TIGER method works as following:

- The first step: for each site in the alignment *S*, the TIGER method partitions sequences into different subsets such that sequences of the same subset have the same state. For example, if the states of six sequences at the site i are "AACGAA", they will be partitioned into $P(i) = \{\{1,2,5,6\},\{3\},\{4\}\}$, i.e., positions 1, 2, 5, and 6 are grouped together because their states at site i are all 'A'.

- The second step: for any pair of sites i and j, the "partition agreement score" $pa(i,j)$ is computed as followings:

$$pa(i,j) = \frac{\sum_{x\in P(j)} a\big(x, P(i)\big)}{|P(j)|}$$

where

$$a\big(x, P(i)\big) = \begin{cases} 1 \; if \; x \subseteq A \text{ for some } A \in P(i) \\ 0 \text{ Otherwise} \end{cases}$$

Note that, $pa(i,j)$ and $pa(j,i)$ are not the same

- The last step: The rate r_i for site i is defined by the mean of all partition agreement scores across all sites:

$$r_i = \frac{\sum_{j\neq i} pa(i,j)}{n-1}$$

The complexity of the TIGER method is $O(n \times l^2)$. Therefore, the TIGER method is only applicable for alignments with few thousands of sites. In other words, it cannot handle genomic datasets with long alignments.

B. The fastTIGER method

The TIGER method calculates the rate of a site by directly comparing its content with the contents of all other sites. The fastTIGER method is designed to avoid expensive computation. The fastTIGER method estimates the similarity between two sequences by comparing their contents over all sites of the alignment. Technically, it summarizes the similarity of sequence pairs into a square matrix $F = \{f(i,j)\}$ of $n \times n$ where $f(i,j)$ is the similarity between two sequences i and j. The evolutionary rate of a site will be calculated from the similarity matrix F. Given an alignment *S*, the fastTIGER method is described as following:

- Step 1: Counting the number of common characters for each pair of sequences. Consider two sequences $S_i = s_{i1}s_{i2}\ldots s_{il}$ and $S_j = s_{j1}s_{j2}\ldots s_{jl}$, we denote $f(i,j)$ is the number of sites k where $s_{ik} = s_{jk}$.

$$f(i,j) = \sum_{k=1}^{l} a_{ijk} \text{ where } a_{ijk} = \begin{cases} 1 & if \; s_{ik} = s_{jk} \\ 0 & if \; s_{ik} \neq s_{jk} \end{cases}$$

The value $f(i,j)$ indicates the similarity between two sequences S_i and S_j, i.e., the greater value means the more similarity between S_i and S_j. Obviously, $f(i,j) = f(j,i)$. The symmetric matrix F represents the similarity between all pair of sequences.

- Step 2: Calculating the rate for each site. Let $C_k = c_{k1}c_{k2}\ldots c_{kn}$ be the states of n sequences at site k. For each site k, we calculate the rate r_k for the site k as following:

$$r_k = \frac{\sum_{c_{ki}=c_{kj}} f(i,j)}{\sum_{i,j=1}^{n} f(i,j)}$$

The rate r_k ranges from 0 to 1. If a site contains more sequence pairs that have the same states at many sites, it will have a slower evolutionary rate.

For example, given an alignment S of four sequences with eight sites as below:

A	A	C	T	G	A	T	C
A	T	C	G	G	G	-	C
T	A	C	G	G	G	A	G
A	-	T	G	G	A	G	C

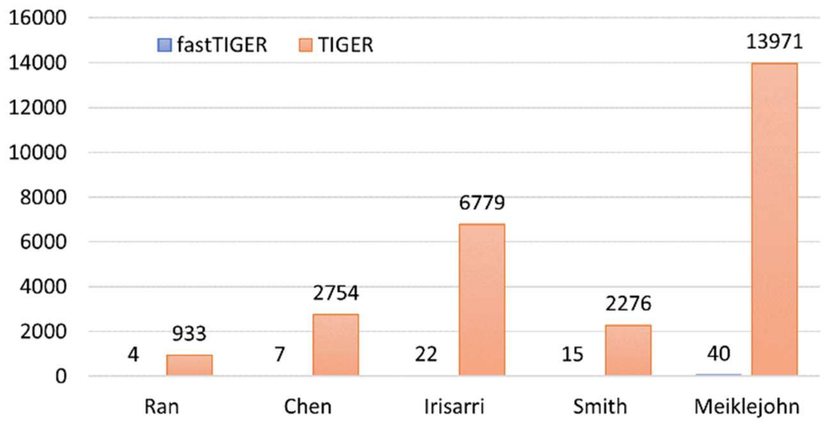

Fig. 2. Time to computing evolutionary rates of fastTIGER and TIGER (unit: second). fastTIGER is our approximate evolutionary rate

We first build the similarity matrix $F = \{f(i,j)\}$ for all sequence pairs:

	4	3	4
4		4	4
3	4		2
4	4	2	

$$\Rightarrow \sum_{1\le i<j\le 4}^{4} f(i,j) = 21$$

Then we compute the rate for each site as following:

The 1^{st} site "AATA" has same state 'A' at position 1, 2 and 4 $\Rightarrow r_1 = \frac{f(1,2)+f(1,4)+f(2,4)}{21} = \frac{12}{21}$

The 2^{nd} site "ATA-" has sane state 'A' at position 1 and 3 $\Rightarrow r_2 = \frac{f(1,3)}{21} = \frac{3}{21}$

The 3^{rd} site "CCCT" has same state 'C' at position 1, 2 and 3 $\Rightarrow r_3 = \frac{f(1,2)+f(1,3)+f(2,3)}{21} = \frac{11}{21}$

Similarly, the rates of the remaining sites are:

$r_4 = \frac{f(2,3)+f(2,4)+f(3,4)}{21} = \frac{10}{21}$;

$r_5 = \frac{f(1,2)+f(1,3)+f(1,4)+f(2,3)+f(2,4)+f(3,4)}{21} = \frac{21}{21}$;

$r_6 = \frac{f(1,4)+f(2,3)}{21} = \frac{8}{21}$;

$r_7 = \frac{0}{21}$;

and $r_8 = \frac{f(1,2)+f(1,4)+f(2,4)}{21} = \frac{12}{21}$.

If a site consists of many different states, the rate approaching 0, and the site is considered to evolve rapidly. Oppositely, a site is considered to evolve slowly if its rate approaching 1 (if most sequences at the site have the same state). A site is called an invariant site if all states at the site are the same (its rate is 1)

The complexity of the fastTIGER method is $O(n \times n \times l)$. Therefore, it can estimate the site rates for long alignments and is applicable for analyzing genomic datasets.

III. Experiments and Results

TABLE I. Six alignments used in experiments

Alignments	Clade	#Taxa	#Loci	#Sites
Chen [15]	Jawed vertebrates	58	40	15278
Irisarri [16]	Jawed vertebrates	100	40	15698
Ran [17]	Seed Plants	38	40	14749
Meiklejohn [18]	Gallopheasants	18	1501	614159
Smith [19]	Plain xenops	8	1366	825804
McCormack [20]	Birds	33	1541	539526

A. Materials

We tested both TIGER and fastTIGER methods on six big alignments including three real protein alignments and three real nucleotide alignments (see TABLE I.). The protein alignments were concatenated from 40 loci which were randomly selected from original alignments (each protein alignment contains around 15 thousand sites). The nucleotide alignments consist of more than 500 thousand sites, i.e., much longer than the nucleotide alignments.

TABLE II. Correlation of two rates over five alignments

Alignments	Correlation
Chen	84.07%
Irisarri	85.00%
Ran	80.48%
Meiklejohn	89.15%
Smith	92.52%

B. Results

First, we examined the running time of both TIGER and fastTIGER methods on the six alignments. The fastTIGER took 110 seconds to calculate rates for the McComack alignment while the TIGER method could not finish the task after 3 days, so we terminated the TIGER method and compared two methods on five remaining alignments. This indicates that the TIGER rate is not applicable for large genomic alignments. Figure 2 shows that the fastTIGER

TABLE III. THE (A) AICC AND (B) BIC SCORES OF RATEPARTITION AND MPARTITION USING FASTTIGER AND TIGER METHODS (I.E., CASES THAT FASTTIGER PRODUCED BETTER TREES THAN TIGER ARE EMPHASIZED IN BOLD).

A	RatePartition		mPartition	
	fastTIGER	*TIGER*	*fastTIGER*	*TIGER*
Chen	**717441**	719504	**710049**	713599
Irisarri	**778724**	782565	772052	765490
Ran	601247	596778	596428	593330
Meiklejohn	2391734	2388237	2149169	2114640
Smith	**2323556**	2324968	**1679620**	2276615

B	RatePartition		mPartition	
	fastTIGER	*TIGER*	*fastTIGER*	*TIGER*
Chen	**718431**	597733	**712780**	715922
Irisarri	**780631**	784897	774482	768843
Ran	602339	596778	598816	595342
Meiklejohn	2392515	2389925	2157040	2123067
Smith	**2324033**	2325481	**1684606**	2279986

method is several orders of magnitude faster than the TIGER method. For example, the fastTIGER methods required only 40 seconds to compute the rates for the Meiklejohn alignment while the TIGER method took 13971 seconds to complete the task (i.e., the TIGER method was about 350 times slower than the fastTIGER method on the Meiklejohn alignment). The running time of the fastTIGER method increases linearly with the length of the alignment, therefore, it is applicable to analyze whole genome datasets.

Second, we calculated the correlations between the rates estimated from the TIGER method and the ones estimated from the fastTIGER method. TABLE II. shows a reasonable correlation between rates estimated from the two methods. For example, the correlation between rates estimated from the TIGER and fastTIGER methods on the Smith alignment is over 92%.

For testing the applicability of evolutionary rates estimated from the fastTIGER method in evolutionary studies, we employed the rates calculated by the fastTIGER and TIGER methods to partition alignments by RatePartition [13] and mPartition [14] partitioning methods. A partitioning method classifies sites into disjoint subsets (called a partitioning scheme) containing sites so that each site in an alignment belongs to one and only one subset in the partitioning scheme. Sites belonging to the same subset are considered to evolve under the same evolutionary pattern. We assessed the quality of a partitioning scheme by reconstructing the maximum likelihood phylogeny from the partitioning scheme (we used IQTREE software [21] to build the maximum likelihood tree). The higher likelihood phylogeny indicates the better partitioning scheme. Technically, we used the AICc [22] and BIC [23] criteria to compare maximum likelihood trees to compromise the difference in the number of free parameters. Note that the smaller AICc (BIC) the better phylogenies.

TABLE III. shows that the fastTIGER and TIGER methods help construct equally good maximum likelihood phylogenies. The results from the RatePartiton method indicate that the fastTIGER method was better than the TIGER method in three alignments (Chen, Irisarri, and Smith) and worse than in two alignments (Ran and Meiklejohn). According to the mPartition method, the fastTIGER method was better than the TIGER in two alignments (Chen and Smith) but worse than in three alignments (Irisarri, Ran, and Meiklejohn). We note that in most tested cases, the AIC/BIC difference between trees constructed with the fastTIGER and TIGER methods is small, except for the Smith alignment where the mPartition method fastTIGER (AICc = 1679620) is considerably better than the mPartition method with TIGER (AICc = 2276615).

IV. DISCUSSION AND CONCLUSIONS

Nowadays, we usually analyze datasets with multiple genes or even whole genomes to study the evolutionary relationships among species. The heterogeneity of evolutionary processes among sites of an alignment must be properly handled to obtain correct analyses. Calculating proper rates for sites provides several advances: 1) site rates can be used to remove outliner sites with too fast evolutionary rates to reduce the large rate variation in sites to clean noise in the data; 2) we can use the site rates to partition sites into subsets to handle the rate heterogeneity in the data such that sites in the same subset evolve under the same evolutionary model.

Although different methods have been proposed for estimating site rates, their limitations are that they are either tree-based methods that might be biased due to the incorrect tree structures or unable to handle large datasets. In this study, we present the fastTIGER method to estimate evolutionary rates for sites by considering the similarity between sites. The fastTIGER method is tree independent and can run remarkably fast even for large genomic alignments. Experiments showed that the rates estimated from the fastTIGER method were as good as ones estimated from the TIGER method. The fastTIGER method enhances us to study evolutionary relationships among species using their genomic data.

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